

An AR-Assisted Deep Learning-Based Approach for Automatic Inspection of Aviation Connectors

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Abstract—The mismatched pins inspection of the complex aviation connector is a critical process to ensure the correct wiring harness assembly, of which the existing manual operation is error-prone and time-consuming. Aiming to fill this gap, this article proposes an augmented reality (AR)-assisted deep learning-based approach to tackle three major challenges in the aviation connector inspection, including the small pins detection, multipins sequencing, and mismatched pins visualization. First, the proposed spatial-attention pyramid network approach extracts the image features in multilayers and searches for their spatial relationships among the images. Second, based on the cluster-generation sequencing algorithm, these detected pins are clustered into annuluses of expected layers and numbered according to their polar angles. Finally, the AR glass as the inspection visualization platform, highlights the mismatched pins in the augmented interface to warn the operators automatically. Compared with the other existing methodologies, the experimental result shows that the proposed approach can achieve better performance accuracy and support the operator's inspection process efficiently.

Index Terms—Augmented reality (AR), aviation connector, deep learning, industrial inspection, spatial-attention pyramid network.

I. INTRODUCTION

WITH digitalization services permeating the manufacturing industries, the modern industries are characterized by the increased automatization and reconfigurability [1]. In this context, the digital innovation enables the flexible planning and managing of the production process cost effectively and, hence, is capable of meeting the customized requirements, achieving improved quality, and reduced time to market [2].

Manuscript received December 10, 2019; revised April 24, 2020 and May 22, 2020; accepted June 4, 2020. Date of publication June 9, 2020; date of current version November 20, 2020. This research was supported in part by the Civil Airplane Technology Development Program under Grant MJ-2017-G-70, in part by the Beijing Key Laboratory of Digital Design and Manufacturing Project, and in part by the Departmental General Research Fund (G-UAHH) from the Hong Kong Polytechnic University, Hong Kong. Paper no. TII-19-5279. (*Corresponding author: Lianyu Zheng.*)

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Digital Object Identifier 10.1109/TII.2020.3000870

Industrial inspection has always been playing a critical role to enable the final production quality of the aviation industry due to its complex structure design and stringent quality requirements. Nevertheless, the conventional manual operation is error-prone and time-consuming, where introducing the novel digital approaches have attracted many concerns during the past decade. For instance, Boeing introduced a fault isolation system [3] to train the operators to quickly learn troubleshooting techniques of the engine air bleed system. In the avionics industry, a new robotic endoscope [4] was introduced to automatically inspect the different residuals (e.g., sand and metallic dust) inside the oil ducts of the metallic gearbox. Despite these achievements, however, due to the complex scenarios in aviation, there still lies several challenges to realize the automatic industrial inspection.

One major challenge is that the existing industrial inspection systems, such as industrial robots and specialized vision equipment [5], fail to present good flexibility and adaptability in those narrow spaces of aerospace products. To address this, the mobile and wearable augmented reality (AR) systems allow coinspection [6] between the remotely connected customers and experts inside the complex structures of products. With the characteristic of the mobility and intuitionistic visualization, the development of an AR-assisted environment provides the possibility of automatic inspection at the manufacturing sites.

Nevertheless, since the AR-based implementation collects the vision information suffering from the complex onsite background noises, most existing visual inspection approaches cannot guarantee an effective performance of the detection accuracy. The conventional methods, such as template matching [7] and filter-based feature selection [8], perform well on the specialized vision-inspection platforms [9] with stable environmental conditions but show a poor performance on dynamic industrial scenarios. To overcome this challenge, a recent work of the deep learning-based inspection has shown unique advantages to identify the product status and quality in a complex manufacturing environment [10]. In this context, the powerful deep neural networks can achieve reliable results for the industrial inspection of the defects from the visual information with heavily textured backgrounds [11]. Despite its good performance, to the authors' knowledge, few works leverage the deep learning technologies and AR-based tools together to conduct the autonomous industrial inspection for the components inside the aerospace products with narrow spaces and complex structures. Most existing works of the industrial inspection only look for the start-of-art performance accuracy by introducing the high-precision visual sensors and optical systems, which fail to provide the

flexibility and adaptability services for inspection tasks that may be dynamically changed.

Aiming to fill this research gap, this article proposes a deep learning-based inspection approach with the AR-assisted visualization platform to automatically detect the mismatched pins of the aviation connector in the wiring harness (ACWH). The rest of this article is organized as follows. Section II introduces the related works of the AR-assisted and deep learning-based technologies in manufacturing. Section III illustrates the prerequisite information of ACWH that are collected from the onsite workplace. The flowchart of the ACWH inspection process is described in Section IV, where the proposed spatial-attention pyramid network and postprocessing algorithm are presented in detail. Section V compares the proposed deep learning model with the existing start-of-art detectors to evaluate the performance. Meanwhile, the processing results of the major steps of the automatic inspection method are presented as well. Section VI discusses the experimental results of the inspection work and other potential implementations. Finally, Section VII concludes this article.

II. RELATED WORKS

This section summarizes the fundamentals of the ACWH inspection and gives a comprehensive review of the recent development and implementation of the AR systems and deep learning approaches to automate the industrial inspection.

A. Fundamentals of the ACWH Inspection

The industrial inspection of ACWH is a very critical operation to guarantee the transmission of the electrical signals all around the system of products. The field of the ACWH inspection can be mostly categorized into two subfields: position inspection and inspection of mismatched pins.

The main idea behind the former is to diagnose the position error of pins in the stage of mating connectors [12]. With force sensor signals, a robotic system was applied to detect and recovery these errors [13]. Based on machine vision, a measurement system consisting of the charge-coupled device cameras was developed to detect the pins of electric connectors [14]. The system can be applied to replace the manual inspection, which is inefficient and susceptible to subjective factors. To verify the position of the pins both in height and in the 2-D plane where they lie, a stereo vision system [15] was designed to measure the position of the pins that are present in various shapes. Then, the deep neural network and pattern matching based on the prior knowledge registration were introduced to a 3-D-vision measurement platform, which can detect the abnormal positions of the pins for a total of 33 types of connectors [16].

The objective of the latter one is to ensure the correct wiring harness assembly. The wiring harness assembly for the aviation connectors is a complicated operation [18], including the recognition of wires, the connection of wires and pins based on the identified text, and the mismatched pins inspection, which presents the consistency between the pins and wires. The operators are only allowed to connect the consistent optical fiber

cords at the connector for the distribution work in central offices [17]. Similarly, not all the pins of the connectors are allowed to be connected with the wires, especially among the aerospace products. Nevertheless, scarcely any existing work provides an autonomous inspection for the mismatched pins of ACWH. A critical challenge that hinders the inspection task is that the connectors are fixed with the wiring harness in the narrow space of these aerospace products.

B. AR Technologies and Tools for Manufacturing

In the era of advanced manufacturing, the AR technologies characterized by digital information are increasingly applied in the production of complex products to achieve visual guidance, interactive collaboration, information management, etc. Based on the 3-D object registration methods, an assembly training and guidance system [19] was developed to render the assembly planning, status of the inspection, and quality evaluation of the products on the model of the cabin parts in an AR environment.

To enhance the interaction between human cognition and manual operation, an AR guidance system [20] was established to render appropriate information on a bare-hand interface by evaluating the users' behavioral experiences during the different phases of tasks. Aiming to develop the AR systems in an interaction-free manner, a gesture-based behavior recognition method [21] was proposed via speeded up robust features (SURF) feature matching. With a head-worn system, the operators can also share onsite information with remote managers. Normally, most existing AR-based platforms are passive due to the lack of feedback. To aware of the operation steps, a haptic module [22] was introduced to track motions of the user's wrist.

For the highly customized production, the AR glasses are employed to transfer the dynamically changing information and instructions through a cloud environment to station operators [23]. With tasks assigning to the production schedules, the intuitionistic instructions are transmitted to the human operators on demand, increasing its flexibility. Therefore, the AR technologies and tools can provide a flexible, adaptable, and visualizing platform for the automatic inspection in manufacturing.

C. Deep Learning Approach for Industrial Inspection

One of the major stages for the industrial inspection is to detect potential objects from the visual information. Recently, the deep learning-based object detectors have shown unique advantages on the large-scale image databases, such as Pascal VOC [24] and MS COCO [25]. A Faster R-CNN [26] was proposed to detect the objects in images with two stages, including the candidate extraction and verification. For the objects with varied scales, the feature pyramid network (FPN) [27] was introduced to extract the intrinsic information from multilevel feature maps. To avoid the time-consuming process caused by the two-stage detection, a single-stage detector YOLO [28] was proposed to meet the real-time need. The major difference is that YOLO divides the entire image into multiple grids, which are used to directly identify the objects' positions and types. But YOLO damages the performance to some extent. The successive versions, i.e., YOLO v2 [29] and YOLO v3 [30], greatly improve performance

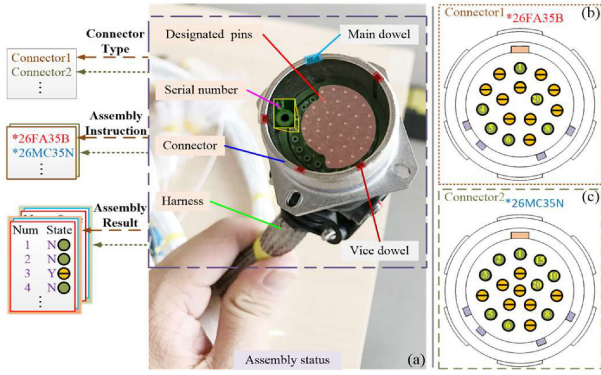


Fig. 1. Structural information of ACWH.

accuracy and remain efficient. Another one-stage detector SSD [31] achieves a higher detection performance for the objects of small sizes, as it adopts the trick of the anchor proposed in Faster R-CNN. To achieve competitive performance equal to the two-stage detectors, the focal loss was applied to the one-stage detector RetinaNet [32] to tackle the problem of the extreme class imbalance during the training.

With the vast amounts of data generating in smart factories, the deep learning-based methods facilitate the development of the analytic and inferential capability for the industrial inspection. For instance, a fog computing-based inspection system [33] was introduced to detect the defects from images of the production line. The system can cope with extremely large data in real time by offloading the computation burden from the central server to fog nodes. For the defect detection of the carbon fiber reinforced polymer (CFRP), an unsupervised machine learning method [34] was introduced to extract the weaker deboned defect from the noise and background. Moreover, with the ultrasonic thermography visualizing the barely visible impact damage, a damage inspection method based on manifold learning [35] was proposed to enhance the detectability for the CFRP damage.

From the above-mentioned literature, one can find that most existing studies implement either AR-assisted technologies or deep learning approaches for industrial inspection, respectively. Nevertheless, few are concerned about the integration of both, let alone to facilitate the flexible and adaptable inspections in the complex aviation scenarios (e.g., ACWH inspection). Motivated by this, a novel approach should be provided.

III. PREREQUISITE INFORMATION OF ACWH

This section illustrates the basic information of ACWH and its image data acquisition and analysis method, as the prerequisite information to achieve the automatic inspection.

A. Structural Information of ACWH

As shown in Fig. 1(a), the structural information of ACWH includes the types of connectors, assembly instructions, and assembly results, and it is normally stored in the digital forms (XML files) in today's aviation industry. Each pin of the connector owns its specific serial number. Outside of all the pins, there is one main dowel and four vice dowels distributed, and the main



Fig. 2. Image acquisition and categories of pins and dowels.

dowel indicates the position of a pin of the initial serial number (e.g., No. 1). For the different types of connectors, assembly instructions, and assembly results, assigning a series of the pins to be connected with the specific wires should be provided during the assembly, as illustrated in Fig. 1(b)–(c). In this context, the inspection of mismatched pins is an indispensable step to guarantee the correct assembly.

B. Collection of the ACWH Image Dataset

To train an effective classifier for the connectors and pins, the training dataset should contain enough image samples and all potential scenarios. For the dataset acquisition, 1724 images are captured from the manufacturing factories by the high-definition digital camera of AR glasses. These images are shot from different angles and contain complex industrial backgrounds, as shown in the upper part of Fig. 2. Meanwhile, these images contain the imaging effects caused by the change of the light intensity to ensure the universality of the image dataset. As shown in the lower part of Fig. 2, pins and dowels are classified into eight types to indicate the actual assembly status. J1 and J2 represent the pins with wires or without wires for the female connectors, respectively, while P1 and P2 are the related examples for male connectors. The dowels on the female connectors are denoted by DP1 and DP2, with the latter signature representing the main dowel. Similarly, DH1 and DH2 carry the same implication for the male connectors. As observed in detail, the dowels suffer contradictory features (e.g., colors and shapes). The pins of the female connectors have only a few characteristics and graphic features, while the pins of the male connectors are extremely small.

C. Statistical Analysis of the ACWH Images Dataset

To analyze the geometric sizes of pins and dowels, connector regions are scaled to the standard images of 1024×1024 pixels. Fig. 3 shows the relationship between the geometric lengths of different pins and their amounts in the dataset. It can be found that the pins and dowels of the connectors present a huge variation range of sizes, of which the maximum length

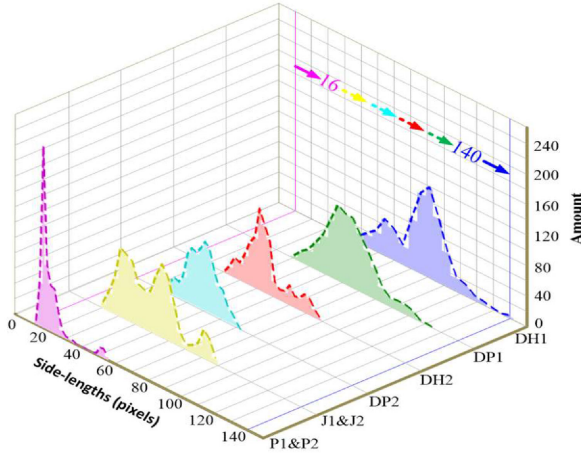


Fig. 3. Statistics analysis of the pins and dowels dataset.

is 140 pixels and the minimum value is only 16 pixels. Also, there is a large disparity of sizes between the largest and the smallest sample of each category.

IV. PROPOSED INSPECTION APPROACH

This article proposes an AR-assisted deep learning-based inspection approach to automatically inspect the mismatch pins of ACWH. With the support of AR glasses, the visualized information can be transmitted to the onsite operators, where the proposed four-step inspection flowchart is depicted in Fig. 4.

- Step 1:* The structural information (XML files) and wireframe images (e.g., CAD models) stored in the industrial systems are shared with AR glasses. The onsite connector image is captured by the digital camera of AR glasses.
- Step 2:* With computing resources from digital manufacturing, the connector detection model is employed to locate the connector region and recognize its type from the preprocessing images (e.g., normalizing and channel modifying). The connector region is cropped from the input as a separate image. A pin detector, i.e., spatial-attention pyramid network is applied to locate the pins and dowels from the cropped image and classify their categories, e.g., *P1* or *P2*.
- Step 3:* A pin-distribution recovering approach is employed to obtain its original distribution. The pins are then sorted by the cluster-generation sequencing algorithm with serial numbers.
- Step 4:* Since the assembly results from structural information are the ground truth (GT), the mismatched pins can be identified by comparing the assembly status with the GT in order. The AR glass delivers warning information to the operators with an intuitionistic interface to highlight those mismatched pins.

A. Spatial-Attention Pyramid Network

An object detector, RetinaNet, is utilized to acquire the coordinates and types of connectors from the images that contain the

complex manufacturing background. According to the output coordinates from RetinaNet, the connector region is cut down as a single image to detect the pins and dowels thereafter.

To detect the pins and dowels of extremely small sizes and various scales, a network of the FPN architecture is built to extract both large features and small features from the images by integrating the feature information between the high-level layers and shallow layers. Meanwhile, BiLSTM [36] is adopted to replace the convolution operation in deep-level layers so as to extract the spatial rules of the pins and dowels distributed on connectors. In this way, the proposed spatial-attention pyramid network not only focuses on the graphic features of the images (e.g., edge, color, and corner) in the various field of views but also searches for the spatial relationships of extracted features.

The architecture of the spatial-attention pyramid network is shown in Fig. 5. The model mainly consists of the feature extractor stem, feature pyramid net, and output layer. ResNet50 is employed as the backbone of the feature extractor branch. In the feature pyramid branch, the features of high-level layers and low-level layers are gradually integrated as follows:

$$b_i = \begin{cases} \text{conv}_{1 \times 1}(f_i) & , \text{ if } i \leq 2 \\ B_C(f_i) & , \text{ otherwise} \end{cases} \quad (1)$$

$$g_i = \begin{cases} \text{add}(b_i, \text{upsample}(b_{i+1})) & , \text{ if } i = 3 \\ \text{add}(b_i, \text{upsample}(g_{i+1})) & , \text{ otherwise} \end{cases} \quad (2)$$

where the B_C layer contains the BiLSTM components, as the detailed structure is shown in Fig. 6. For each pixel of feature maps, a sliding window (3×3 pixels) is introduced to gather the values of this pixel and surrounding eight pixels, and these nine values are output to the final channel of feature maps. Hence, this processing, named extract patch, increases the information for the feature map by expanding its dimension from (B, H, W, C) to $(B, H, W, 3 \times 3 \times C)$ without new training parameters introducing into the detection model. Hence, it does not aggravate the difficulty of training. To meet the requirements of the input data of BiLSTM, the feature maps that output a form extracting patch are reshaped to $(B \times H, W, 3 \times 3 \times C)$ as the sequence data. The sequential data are sent to BiLSTM to generate the feature maps of $(B \times H, W, C)$, which contains the information of the position correlations of pins and dowels. Finally, the output feature map is recovered to (B, H, W, C) for afterward deeper layers of the network, where C is defined as the number of the channels of feature maps.

The B_C layer is only applied to the deeper layers of Res5c_relu and Res4f_relu, which are the output from ResNet50. By comparison, the shallow layers are not effective enough to extract the intrinsic information of images. More importantly, the disadvantages of the shallow feature maps, such as large sizes and fewer channels, can lead to a short data sequence, which requires more basic units in the chained BiLSTM components. As these problems make the training process more difficult and even degenerate the accuracy performance of the networks, the B_C layer is not introduced to the low-level feature maps.

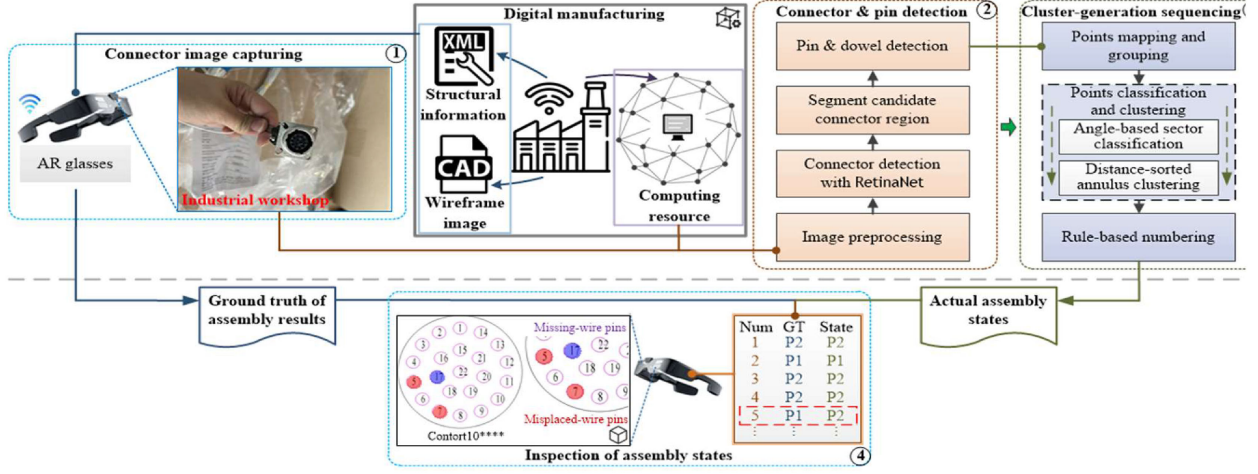


Fig. 4. Flowchart of inspecting mismatched pins.

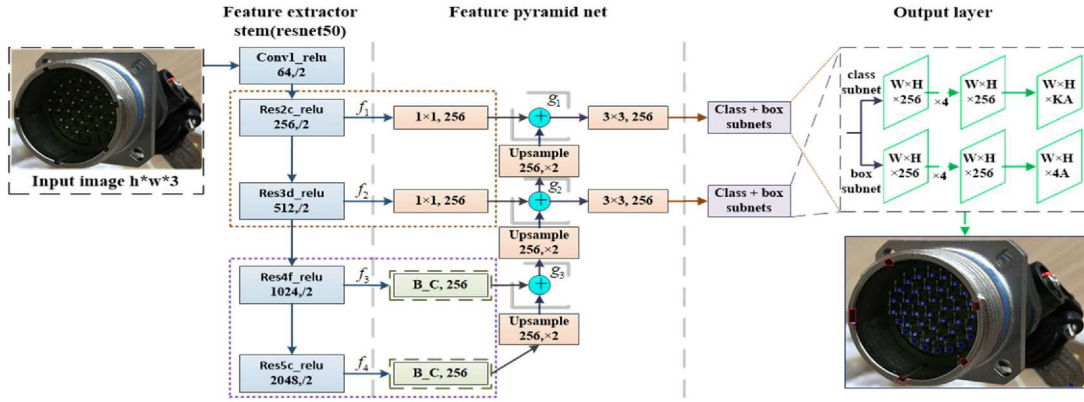


Fig. 5. Architecture of the spatial-attention pyramid network.

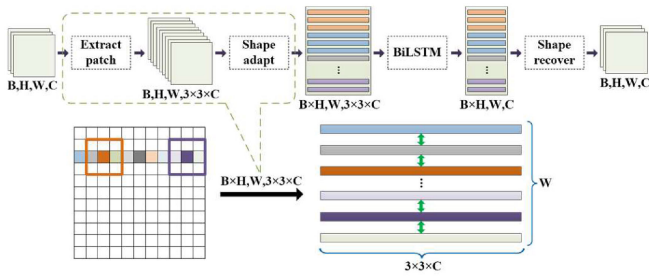


Fig. 6. Spatial-attention layer with the BiLSTM component.

The output layer is simple and efficient, as the subnets of classification and regression are only applied to the final two upper feature maps of the feature pyramid branch. Each pixel (i.e., sliding position) of the feature maps of these two subnets is related to one anchor. More specifically, anchors on the top subnet and the bottom subnet, respectively, represent 16^2 and 32^2 pixels of input images. The values of the sliding steps of anchors are set to 4 and 8 for these two subnets, respectively. Finally, nine

anchors are yield on every sliding position by multiplying three scales (2^0 , $2^{1/3}$, and $2^{2/3}$) and three aspects ratios (2^0 , $2^{1/3}$, and $1+2^{2/3}$) to each anchor. These anchors cover scale ranges from 16 to 82 pixels of input images. Every anchor is assigned to a length K one-hot vector for classification targets and a four-vector for box regression targets. K is the number of the classes of pins and dowels and is set as eight.

In the classification subnet, the probability of object presence at each sliding position is indicated by nine anchors and K object classes. In the end, sigmoid activations are attached to output $9K$ binary predictions for each sliding location.

In the box regression subnet, there are four outputs predicting the relative coordinates v of the bounding box location of one anchor. The relative coordinates v are given by

$$\begin{aligned} v_x &= (c_x - c_x^a) / w^a, \quad v_y = (c_y - c_y^a) / h^a \\ v_w &= \log(w - w^a), \quad v_h = \log(h - h^a) \end{aligned} \quad (3)$$

where $v = \{v_x, v_y, v_w, v_h\}$ are the relative predicted coordinates. $\{c_x^a, c_y^a, w^a, h^a\}$ denote the center coordinates and its width and height of the anchor box, which can be precomputed

from an input image. $\{c_x, c_y, w, h\}$ are the predicted coordinates of the bounding box. To jointly optimize the parameters of the pin detector, two loss functions, L_s^{cl} and L_v^{re} , are introduced to calculate the errors of subnets of the classification and box regression, respectively. The loss is formulated as

$$L = L_s^{cl} + \lambda_{re} L_v^{re} \quad (4)$$

where λ_{re} balances the influence of two losses on the training and is set to one. In the image dataset, there is a large disparity in the number of pins and dowels from different classes. To solve the problem of class imbalance in the classification task during training, L_s^{cl} adopts a focal loss, which is expressed as follows:

$$L_s^{cl} = \begin{cases} -\alpha(1 - \hat{y})^\gamma \log \hat{y}, & \text{if } y = 1 \\ -(1 - \alpha) \hat{y}^\gamma \log (1 - \hat{y}), & \text{otherwise} \end{cases} \quad (5)$$

where y is the label of instances, and α balances the importance of the positive/negative examples (i.e., candidates generated by the network are potential objects or not) and is set to 0.25 in the experiment. The γ trades the weights between the easy/hard examples and enables training to focus on the hard examples (i.e., losses of examples with errors) and is set to 2.

The loss function of regression tasks in the object detection models normally applies the L_n norms, which are defined by the distance between the corners of two boxes. As intersection over union (IoU) is a metric used in the evaluation benchmarks, metrics between the training and testing are not consistent. Another problem is that the L_n norms make the detection performance sensitive to the choice of the IoU threshold, especially the detection model may present instability for the objects of various sizes. Hence, L_v^{re} adopts the form of generalized intersection over union (GIOUS) [37] loss that enables the models to present an invariant accuracy against the different choices of IoU thresholds. With $\{x_1, y_1, x_2, y_2\}$ indicating the corners of the predicted bounding box $\hat{R}$ and $\{x_1^*, y_1^*, x_2^*, y_2^*\}$ denoting the corners of the GT bounding box R^* , L_v^{re} is attained by

$$L_v^{re} = 1 - \left(\left| \hat{R} \cap R^* \right| / \left| \hat{R} \cup R^* \right| - \left(R^C - \left| \hat{R} \cup R^* \right| \right) / R^C \right) \quad (6)$$

where R^C is the area of the smallest enclosing box B^C and the coordinates of B^C are as follows:

$$x_i^c = \min(x_i, x_i^*), y_i^c = \min(y_i, y_i^*), \quad i = 1, 2 \quad (7)$$

$$R^C = (x_2^c - x_1^c) \times (y_2^c - y_1^c). \quad (8)$$

B. Pin-Distribution Recovering and Cluster-Generation Sequencing Algorithm

As the shooting angles and focal distances of cameras are uncontrolled, the pins may be distributed on the irregular annulus in the images, which affect the subsequent work to obtain the serial numbers of pins. A homographic matrix H (i.e., perspective transformation matrix) is utilized to recover the original distribution of pins. The formula used to transform

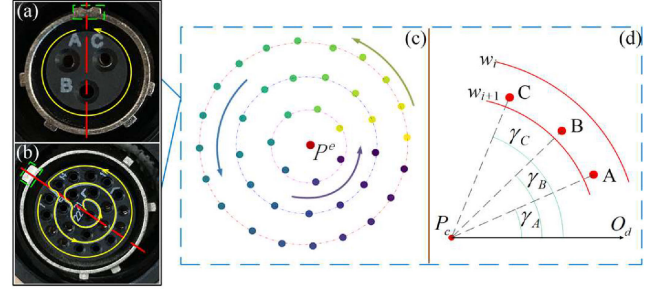


Fig. 7. Spatial rules of the distribution of pins.

two points of (x, y, z) and (x', y', z') with matrix H is given as

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} H_{11} & H_{12} & H_{13} \\ H_{21} & H_{22} & H_{23} \\ H_{31} & H_{32} & H_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}. \quad (9)$$

More than four optimal matching points are required to estimate the parameters of H due to its nine degrees of freedom. One registration image, in which the pins are distributed on the regular multilayer annulus, is selected from every type of the connector image. Then five pairs of the optimal matching points, i.e. one pair of $DP1/DH1$ and four pairs of $DP2/DH2$, are searched over the real-time test image and the registration image to calculate the H . Finally, the coordinates of all pins on the test images are recovered by H to acquire their original distribution on the connector. As illustrated in Fig. 7(a) and (b), the pins of the connectors are distributed on annuli from the outer layer to the inner layer anticlockwise. The number of the annuli layer is determined by the type of a connector. Specifically, the central pin P^e is the last serial number if it exists on connectors. Based on the distribution rule, the strategy of the cluster-generation sequencing algorithm contains the following stages.

- 1) Points mapping and grouping: All detected pins and dowels are mapped to discrete points according to the coordinates of their center. The discrete points related to the dowels are grouping into set d , of which the main dowel (marked as O_d) is a reference of the initial position to sequence all pins. Then these discrete points related to the pins are put into group S . The central point P_c of all points in S is calculated.
- 2) Points classification and clustering: The connector type obtained from the connector detection model presents the information about the number of annuli layer n and whether the given connector has the central pin P^e . With the information of n and P^e , the points in set S are clustered into different annuli from the outer layer to the inner layer. The detailed process is described in Algorithm 1.
- 3) Rue-based numbering: For all points in set S , a polar coordinate system is built with P_c as the pole point and $P_c O_d$ as the polar axis. Based on the polar-angle value γ_i of points, the serial numbers of all pins are acquired by sorting points of every annulus from outside to inside [see Fig. 7(c) and (d)].

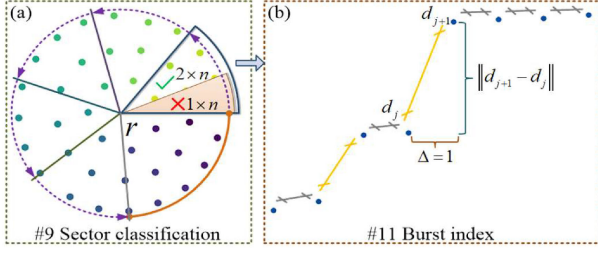


Fig. 8. Point classification and clustering based on angle and distance.

Algorithm 1: Points Classification and Clustering.

Input: Existing flag f_c of P^e , Number of annuluses layer n , Point set of pins S

Output: Point sets in every annulus $\{C_1, \dots, C_n, C_{n+1}\}$

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1: function CLUSTERING ( $f_c, n, S$ )
2:    $P_c(\bar{x}, \bar{y}) \leftarrow \frac{1}{N} \sum_i^N P(x_i, y_i) \forall P_i \in S$ 
3:   if  $f_c$  then
4:     The nearest point to  $P_c$  in  $S$  is moved to  $C_{n+1}$ 
5:   end if
6:    $u = (|S| - 2 * n) / (2 * n) + 1$  // The number of sectors
7:    $r = (|S| - 2 * n) \% (2 * n) + 2 * n$  // The number of points
   in the last sector
8:   Points in  $S$  are sorted based on their polar angles to
   pole point  $P_c$ 
9:   The last  $r$  points in  $S$  is moved to  $C_u$ 
10:  The remaining points in  $S$  are split equally into sets
    $C_1, \dots, C_{u-1}$ 
11:  Points in  $C_1, \dots, C_u$  are sorted by their distance to  $P_c$ 
12:  Calculating the index burst (IB) for points in
    $C_1, \dots, C_u$ 
13:  Points in  $C_1, \dots, C_u$  are split into  $n$  sets  $C_1, \dots, C_n$ 
   with the largest  $n-1$  IB
14:  return  $\{C_1, \dots, C_l, C_{l+1}\}$ 
15: end function

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The processing of points classification and clustering contains two stages, i.e., angle-based classification and distance-sorted annulus clustering. In the angle-based sector classification, the first consideration is that the number of points in each sector should be greater than annuluses' layers n to ensure that every sector contains points (i.e., pins) from all annulus of the connector. Another one is to generate as many sectors as possible to split all points into the fan-shaped distributions of smaller angles, which reduce the difficulty to classify the pins to the different layers of annuluses. Based on these two aims, the number of sectors u is given in Line 6 of the Algorithm 1 and the amount of points r for the last sector is calculated in Line 7. Then, the last r points in set S are moved to the last sector, while the remaining points in S are divided equally to other sectors, as shown in Fig. 8(a).

In the process of the distance-sorted annulus clustering, the burst index [see Fig. 8(b)] that indicates the distance between the points inside each sector to central point P_c is first calculated in Line 12 of the Algorithm 1. Then the largest $n-1$ burst index

in every sector is used to split the points of the corresponding sector into n different annuluses. Finally, the serial numbers of all pins are obtained by sorting these points in the subsequent process of the rule-based numbering. Traversing all numbered pins in order, the mismatched pins are identified by comparing the GT of the assembly result and its actual status.

V. CASE STUDY AND EXPERIMENTAL RESULT

In this section, several comparison experiments are conducted to evaluate the performance of the detection accuracy for the spatial-attention network. Processing results of each step for the inspection of mismatched pins are visualized as well.

A. Pin Detection

There are 11 different types of connectors in the dataset that contains 1346 training images and 378 testing images. RetinaNet is trained as the connector detector to locate the positions of the connector region in an image. The testing images are used to quantify the detection result with the evaluation criterion of the mean average precision (mAP), which is the metric for the object detection in Pascal VOC. The performance of the detection accuracy is presented in Table I.

The connector region is cropped down from the above-mentioned 1346 training images and 378 testing images, respectively, to develop the dataset for the pin-detection model. Finally, 1009 training images and 360 testing images are acquired after removing the images that suffer extremely poor quality, such as the blur and false labels. Similarly, these two groups of images are fed into the spatial-attention pyramid network to train the pin detector and test performance of the trained model, respectively. Without any augmented image, our proposed model is trained with the stochastic gradient descent optimizer over two graphics processing unit (GPUs) (RTX 2080) with a total of two images per minibatch, i.e., each GPU processes one image of 1024×1024 pixels. The model is trained for 80K iterations with an initial learning rate of 0.01, which is then divided by 10 at 40K and again at 60K iterations. The weight decay is set to 0.0001 and the momentum is set to 0.9. The training loss is the sum of the focal loss for the classification and GIOU loss for the box regression.

To verify the performance of the detecting pins and dowels in images, the proposed spatial-attention pyramid network is compared with the first-class 2-D object detectors, including Faster R-CNN, SSD, YOLO v2, RetinaNet, and YOLO v3. These existing object detectors achieve top performance in the most popular public databases, e.g., PASCAL VOC and MS-COCO, etc. Based on the same computing resource (one RTX 2080 GPU, i7-9700k CPU, and 16 GB RAM) and the connector dataset leveraged, Table II presents the performance accuracy and the computation efficiency of our proposed model, and these existing detectors on detecting the pins and dowels from onsite images. Some examples and detection results of the pin detector are illustrated in Fig. 9.

YOLO v3 normally presents a high performance of the detection accuracy for small objects. Compared with YOLO v3, our detector shows a dramatic improvement to detect the pins, such as J1, J2, P1, and P2, which are distributed with some spatial

TABLE I
PERFORMANCE OF CONNECTOR DETECTION ACCURACY

mAP	CONN1	CONN2	CONN3	CONN4	CONN5	CONN6	CONN7	CONN8	CONN9	CONN10	CONN11
0.9961	0.9996	0.9846	0.9925	0.9956	0.9992	0.9979	0.9941	0.9996	0.9977	0.9975	0.9985

TABLE II
PERFORMANCE OF PIN DETECTION

Method	mAP	DH1	DH2	DP1	DP2	J1	J2	P1	P2	time
Faster R-CNN	0.7268	0.9044	0.9078	0.8984	0.9047	0.8325	0.8248	0.2193	0.3225	--
SSD	0.5112	0.8956	0.8595	0.8305	0.5188	0.3432	0.4619	0.0961	0.0625	--
YOLO V2	0.4982	0.8060	0.7849	0.8143	0.7147	0.3430	0.3414	0.0909	0.0909	--
RetinaNet	0.7483	0.9696	0.9892	0.9943	0.9795	0.9870	0.8888	0.0633	0.1152	237ms
YOLO V3	0.9757	0.9935	0.9980	0.9840	0.9653	0.9843	0.9084	0.9834	0.9893	109ms
Ours	0.9900	0.9912	0.9964	0.9829	0.9858	0.9965	0.9709	0.9973	0.9986	318ms

TABLE III
ACCURACY OF MODELS TRAINED BY DIFFERENT PARAMETERS

γ	α	AP50	AP65	AP75
1.0	0.50	0.9480	0.8853	0.7736
1.0	0.25	0.9771	0.9261	0.8054
2.0	0.25	0.9900	0.9772	0.8794
5.0	0.25	0.9891	0.9419	0.8217



Fig. 9. Examples of pin detection.

rules. It means that the BiLSTM structure in the network is activated to extract the spatial features during training and testing. Besides, $J2$ represents the negative hard examples, which are not easy to distinct with $J1$, but our proposed method presents the most significant improvement of the performance on detecting these kinds of confused instances. The reason is that our detector focuses on extracting the features of such negative hard samples during the training with the implementation of the focal loss.

In the MS COCO challenge, the widely reported accuracy measure for the detection, averages mAP, is calculated based on multiple IoU values, e.g., 0.5, 0.6, 0.7, and 0.8, etc. Taking this more reasonable evaluation criterion into consideration, the detection results are quantified with mAP of the different threshold values to verify the advantaged performance of the proposed spatial-attention network. Five different ways of the image expansion are adopted to enhance the original 1009

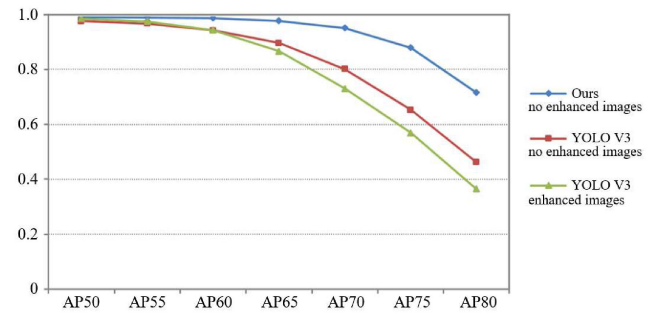


Fig. 10. mAP in multiple IoU thresholds.

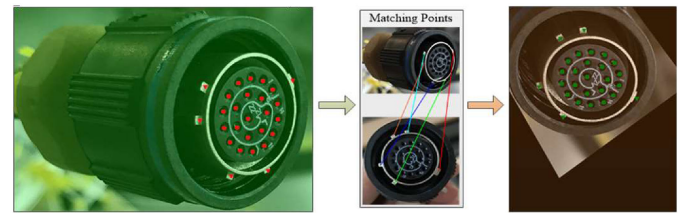


Fig. 11. Pin-distribution recovering.

training images to 6054 images. Our model is trained by the original 1009 images, and the two models of YOLO v3 are trained by 1009 images and enhanced 6054 images, respectively. As shown in Fig. 10, the gap of the detection accuracy between our model and YOLO v3 enlarges as the IoU threshold rises. Our proposed model is less sensitive to the choice of the IoU threshold and maintains a higher detection accuracy even in a large IoU value.

As mentioned earlier, by introducing parameter α and parameter γ in the loss function of the classification task, the proposed model can put attention to hard, misclassified examples during the training. The effects on the accuracy of the models trained by other values of the parameters are illustrated in Table III. To maximize the performance of the model, the parameter α and the parameter γ are set to 0.25 and 2.0, respectively.

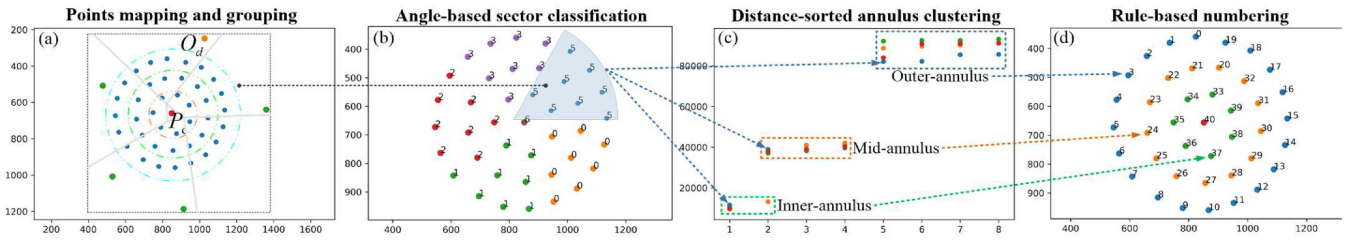


Fig. 12. Visualization of the cluster-generation sequencing algorithm. (a) Points mapping and grouping. (b) Angle-based sector classification. (c) Distance-sorted annulus clustering. (d) Rule-based numbering.

TABLE IV
COSTS OF INSPECTION SYSTEMS

Platform	Industrial Camera-based Inspection System		Robotic System	Ours
	Nanovea ST400 Profilometer	[16]		
Hardware	Color Video Zoom Camera Integration, Adjustable Height experimental platform	Two Basler Industrial Cameras, Light Source, Experimental Platform	Robot, Industrial Cameras, Light Source	HiAR G200
Computing	PC	PC with GTX 1080 GPU	PC	RTX 2080 GPU, i7-9700k
Power				CPU and 16G RAM
Pricing	Over 50,000\$	Around 4,200\$	Over 30,000\$	Around 3,000\$

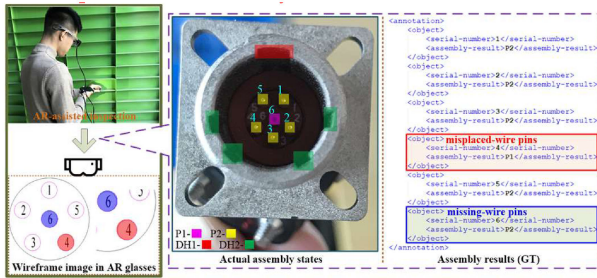


Fig. 13. Mismatched pins inspection with the AR-assisted platform.

B. Visualization for Inspection of Mismatched Pins

With five pairs of the matching points (i.e., dowels of connectors) establishing a homographic matrix H , the pin-distribution recovering algorithm recovers the pin's distribution of every annulus from irregular ellipses to more regular circles (see Fig. 11). It can work against the coordinate deviation of the pins caused by the weird shooting angle of a camera. After the processing of the pin-distribution recovering, the detected pins and dowels are grouped into discrete points and then classified into separated sectors, as illustrated in Fig. 12(a) and (b). Points from every sector are clustered into different annuluses from outside to inside, i.e., outer annulus, mid annulus, and inner annulus, as presented in Fig. 12(c). Finally, 40 points related to the pins obtain their serial numbers correctly by sorting from the outer layer to the inner layer in a counterclockwise direction [see Fig. 12(d)]. The mismatched pins are identified from the inconsistency of comparisons between the GT of the assembly results and the actual assembly status of these numbered pins. The AR glass, as an important interactive device, presents a wireframe image that highlights the mismatched pins to onsite operators. For example, in Fig. 13, the highlight blue pin of No. 6 of connector 5 reminds the operators to install an aviation wire

into this pin, while the red pin of No. 4 denotes a misplaced one to be removed.

With the deployed system, one cycle inspection starts from that a connector image is captured by the AR glasses in real time to the end when the AR glasses display the wireframe image. During the cycle, the majority of the time spends on the calculation of the detection model, where the connectors are detected from images in 224 ms and the pins are recognized from images in 318 ms. Coupled with the rest of the little time consuming, such as image transmission and information interchange, the AR-assisted deep learning-based inspection of the mismatched pins can be performed within 1 s between the manufacturing systems and onsite operators automatically.

VI. DISCUSSION

To detect small pins of various scales of the ACWH, the proposed spatial-attention pyramid network extracts the feature information in a diverse field of view and searches for their spatial relationships. Also, the cluster-generation sequencing algorithm can resist the position deviation of the pins in images to some extent. By integrating on the AR glasses, the automatic inspection method can be applied to any complex industrial scenarios to provide digital inspection services with flexibility and to present an intuitionistic visualization effect. Meanwhile, compared with the other existing industrial camera-based inspection systems [14]–[16] or robotic systems [12], [13], the deployed system has prominent advantages in terms of the competitive price and powerful computation efficiency, as given in Table IV. As computing servers are commonly essential facilities existing in the aviation industries, the only cost of the purchasing hardware for the AR-based inspection system can be less than \$1700. It is envisioned that the proposed approach can also be readily extended to many other applications, such as the target tracking in human-machine collaboration systems. Despite those advantages, the computation efficiency of the

proposed approach (318 ms) is rather slower than the YOLO v3 (109 ms). Therefore, there is still room for further improving its computing efficiency without sacrificing the accuracy.

VII. CONCLUSION

In this article, an automatic aviation connector inspection method was proposed, which was in line with today's digital transformation in the aviation industry. With inferred knowledge transferring from the learning model, an AR glass was leveraged to support the onsite operator's inspection work effortlessly in a visualized manner. To summarize, the main scientific contribution of this research lies in two aspects.

- 1) Proposed a spatial-attention pyramid network combining the architectures of FPN and BiLSTM to address the challenge of detecting small objects of various scales, with exposure to the complex and nature industrial scenarios. Since most machine parts and components have spatial correlations, the proposed model shows great advantages of adaptability and flexibility that can be adopted in various applications.
- 2) Integration of the AR visualization and deep learning technologies to provide automatic inspection services with flexibility and mobility. This article opens a new way to conduct the visualized industrial inspection automatically inside a product regardless of its complex structure or narrow spaces.

Moreover, to validate the proposed approach, an experimental analysis of mismatched pins of 11 different ACWH was conducted and the results show that the pin-detection model achieved the competitive performance with an accuracy of up to 99.00%. Apart from the above-mentioned advantages, several potential future research directions were also highlighted here, which are as follows.

- 1) Optimizing the loss function of the model to improve the performance of the detection accuracy and verifying the feature extraction ability of the model for other important parts of aerospace products.
- 2) Extending the proposed AR-assisted data-driven inspection approach to other industrial scenarios.
- 3) Deploying a cloud-edge computing service platform to facilitate the on-demand inspection service.

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